

# S.N.A.S.H.

## The Auto-Transforming Robot (MON-10)

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An **A**utomated **M**ultimodal **M**obile **M**orphoBot which combines driving and flying for search and rescue, payload delivery, and site inspections, offering extended battery life and multi-terrain adaptability for efficient operations.

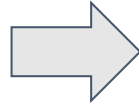
# Status of project

| Key specifications and requirements proposed at beginning of semester | Status at end of semester | Remarks/Justification                             |
|-----------------------------------------------------------------------|---------------------------|---------------------------------------------------|
| Automated                                                             | Complete                  | The bot can transform without manual intervention |
| Multimodal                                                            | Complete                  | Drone and Drive modes demonstrated successfully   |
| Weight < 3kg                                                          | Complete                  | The total weight is 2kg                           |
| Drive speed > 0.5m/s                                                  | Complete                  | The maximum speed of the bot is close to 0.5m/s   |
| Battery Life > 15 mins                                                | Complete                  | Flight time ~2 minutes<br>Drive time ~15 minutes  |
| Morph Time < 1 minute                                                 | Complete                  | Morph time is ~4 seconds                          |

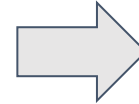
# Photographs of final prototype



Drive Mode



Morph Mode

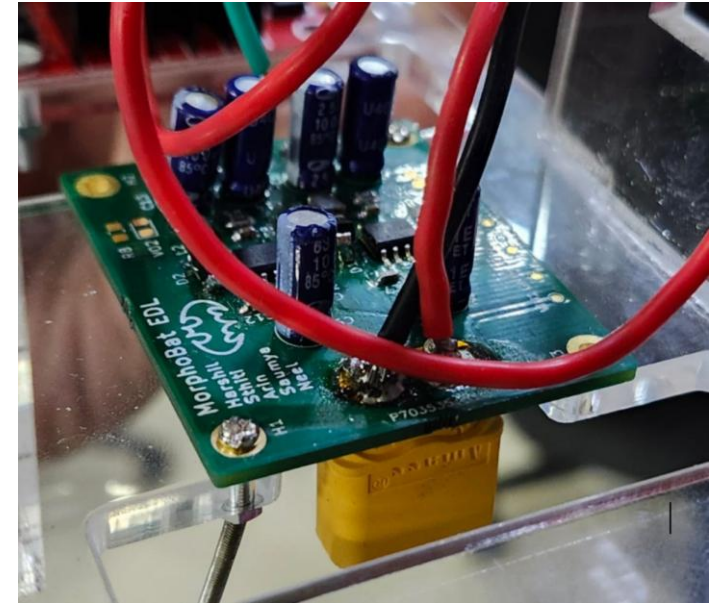
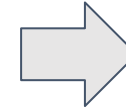
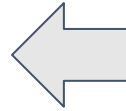


Drone Mode

# Internal and External Components



Wheel Assembly



PCB

# Some Amazing Photos!!

